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RESEARCH ARTICLE

Small-Signal Stability Analysis of Power Systems by Implicit Multilinear Models

CHRISTOPH KAUFMANN[1,2,3], GEORG PANGALOS[1], GERWALD LICHTENBERG[2], AND ORIOL GOMIS-BELLMUNT[3], (Fellow, IEEE)

[1]Application Center for Integration of Local Energy Systems ILES, Fraunhofer Institute for Wind Energy Systems IWES, 27572 Bremerhaven, Germany
[2]Faculty Life Sciences, Hamburg University of Applied Sciences, 21033 Hamburg, Germany
[3]Centre d'Innovació Tecnológica en Convertidors Estàtics i Accionaments, Departament d'Enginyeria Elèctrica, Universitat Politècnica de Catalunya, 08028 Barcelona, Spain

Corresponding author: Christoph Kaufmann (christoph.kaufmann@iwes.fraunhofer.de)

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ABSTRACT This paper proposes a new approach to perform small-signal stability analysis based on linearization of implicit multilinear models. Multilinear models describe the system dynamics by multilinear functions of state, input, and algebraic variables. Using suitable transformations of variables, they can also represent trigonometric functions, which often occur in power systems modeling. This allows tensor representations of grid-following and grid-forming power converters. This paper introduces small-signal stability analysis of equilibrium points based on implicit multilinear models using generalized eigenvalues. The generalized eigenvalues are computed from linear descriptor models of the linearized implicit multilinear model. The proposed approach is tested using an essential power system network example, first by comparing time-domain simulations of the implicit multilinear model with those of the nonlinear model, and second by comparing the generalized eigenvalues with those of the linearized nonlinear model. The results show that the decomposed tensor representation of the implicit multilinear model allows for a faster linearization compared to conventional methods in MATLAB Simulink.

INDEX TERMS Small-signal power system analysis, converter-dominated network, multilinear models, tensors, generalized eigenvalues.

I. INTRODUCTION

Power systems are undergoing major structural changes from fossil fuels to renewables, which have not been seen in the last hundred year with this speed. The paradigm shift from producers and consumers to prosumers, as well as the replacement of large fossil-fueled power plants by numerous renewable energy resources, comes with an increasing complexity of the overall already complex system. This poses challenges in many ways which could be categorized from a viewpoint of systems theory in stability, controllability and observability issues, where any of them are likely to become worse. Moreover, very essential components in power systems cannot be described by linear state-space (LSS) models with the necessary accuracy - especially for transient regimes with nonlinear dynamics.

Methods for small signal stability analysis based on linearizations of nonlinear models are well established in power systems engineering, and also by practitioners [1], [2]. But they become more and more involved with an increasing number of relevant nonlinear components with continuous-valued and discrete-valued dynamics [3], [4]. This paper proposes a new method which helps to cope with these challenges by splitting the linearization process into two steps. First, an offline step, which results in an intermediate multilinear model. Second, a linearization step which can

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be computed very efficiently, and thus even online, because of the tensor parameter structure of the multilinear model obtained after the first step [5].

Consider the calculation of the active power in a three-phase system in *abc*-coordinates

$$p(t) = v_a(t)i_a(t) + v_b(t)i_b(t) + v_c(t)i_c(t), \tag{1}$$

where $v(t)$ and $i(t)$ are the voltages and the currents of the phases. It is a fundamental component in power system modeling, yet it cannot be represented by a LSS model but needs to be approximated. There is new modeling class called multilinear time-invariant (MTI) models [6], [7] that can model (1) exactly, because the right-hand side of (1) is in fact a multilinear function, i.e., it is linear if all but one variable are held constant [8].

Multilinear models can also represent discrete dynamics besides multilinear and linear ones [6], [7]. With the help of Boolean variables, MTI models can represent saturation as it was shown in [7]. Therefore, multilinear models can describe more complex dynamics than linear models. As a result, numerical multilinear approximations of nonlinear state space models can lead to better approximation than a linear one as it was shown for a constant stirred tank reactor in [9], or for an electrolyzer in [7].

MTI models can be represented as tensors, which are k-dimensional arrays and can bring several benefits to modeling of dynamical systems, like the matrix representation does for LSS models. Using tensors to represent MTI models can enable efficient representation. For example, in [7], a canonical polyadic (CP) tensor decomposition method [10] was used to reduce the memory requirement of the MTI model of an electrolyzer by more than 43 % while meeting the accuracy requirement. In [11], a reduction of the number of elements of general CP-decomposed MTI models by almost half were achieved by normalization, called canonical polyadic norm-1 normalized (CPN1).

Mathematical operations can take advantage of the structure of the decomposed tensor of the MTI model. In [5], the CPN1 format was leveraged to develop an efficient linearization method that achieved a gain of computational efficiency for the Jacobian by two orders of magnitude. These specialized methods then can be used in controller design, such as feedback linearization [12], or model-predictive control [13].

MTI modeling has been an active field of research for the last decade with remaining research gaps, for instance, not all methodologies utilize the decomposed tensor structure yet. For example, for time-domain simulation of continuous-time implicit MTI models, existing numerical integration algorithms are used that do not utilize the decomposed tensor structure [14], [15]. For some application fields like heating systems, MTI models have been proven to be a suitable model class [13], [16], [17]. Power system models, however, utilize some nonlinear functions [18], e.g., trigonometric functions, for which a symbolic multilinearization needs to be found, as it is preferable over multilinear approximation [9].

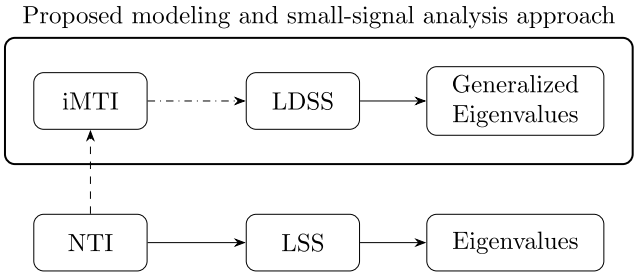


FIGURE 1. Comparison of the steps for small-signal stability analysis using iMTI models and NTI models.

The modeling of power systems as MTI models was first done in [14], where a *pq* open-loop controlled photovoltaic inverter in $\alpha\beta$-coordinates was modeled. In [15], a larger converter-based microgrid using a droop-based control was modeled in *dq*-frame, where a change of variable was used to model the rotational matrix. These works focused on the modeling and simulation of power systems using MTI models. This paper investigates how MTI models can be used to perform small-signal stability analysis of converter-dominated power systems. This work is built upon the ongoing research and extends it by the following contributions:

- Exact symbolic multilinearization of relevant non-multilinear functions for the modeling of grid-following and grid-forming power converters.
- Extending the efficient linearization procedure of explicit multilinear time-invariant (eMTI) models shown in [5] to implicit multilinear time-invariant (iMTI) models leading to local approximations by linear descriptor state-space (LDSS) models.
- Applying the small-signal stability analysis method for models described by nonlinear differential-algebraic equations (DAEs) using generalized eigenvalues [19] to the model class of implicit multilinear time-invariant models.

The proposed methodology will be tested on a converter-dominated network consisting of a grid-forming, and a grid-following converter, connected to a load, and a grid, where load and voltage steps are applied. It will be shown that the proposed methodology allows for an efficient small-signal stability analysis of power systems compared to standard tools. Fig. 1 compares the steps to perform small-signal stability analysis using nonlinear time-invariant (NTI) models in the bottom-half of the figure, and the proposed modeling and small-signal analysis approach in the upper-half of Fig. 1.

The paper is organized as follows. Section II describes the class of iMTI models and its representation as decomposed tensors. Exact multilinear modeling of some nonlinear functions encountered in power converters is discussed in Section III corresponding to the dashed line in Fig. 1. Then Section IV shows the efficient computation of the Jacobian of iMTI models, out of which an LDSS model is

extracted corresponding to the dash-dotted line in Fig. 1. From the LDSS model, the generalized eigenvalues are computed as presented in Section V. Section VI outlines the model used in Section VII, for the comparison of the time-domain simulation and small-signal analysis using Matlab Simulink [20], where a comparison of the computational time of the linearization and the computation of the generalized eigenvalues is provided. Section VIII draws conclusion of the work and defines the next steps.

II. IMPLICIT MULTILINEAR MODELS AND TENSOR REPRESENTATION

MTI models were introduced as in [6] as state space models with multilinear functions as right-hand sides. These MTI state-space models are also referred to as eMTI models because their dynamic behavior is described by ordinary differential equations (ODEs). If the ODEs are restricted by a set of algebraic equations, meaning the model is described by DAEs, the semi-explicit multilinear time-invariant (sMTI) model class [14] can be used. Writing the DAEs fully implicitly leads to the iMTI model class [7].

In this paper, the set of real numbers is denoted by $\mathbb{R}$, where a scalar parameter is indicated by a non-bold face $x \in \mathbb{R}$, lower case bold face is used to indicate a column vector $\mathbf{x} \in \mathbb{R}^n$, upper case bold face for matrices with $\mathbf{M} \in \mathbb{R}^{n\times m}$, and upper case non-bold face for a k-th order tensor $\mathsf{T} \in \mathbb{R}^{I_1\times I_2\times\ldots\times I_k}$.

In the subsequent section, the representation of iMTI models as decomposed tensors is briefly presented starting with the full tensor representation.

To formulate an iMTI state-space model, the monomial tensor is defined at first.

Definition 2.1: [6], [7], [10] A *monomial tensor* is defined as

$$\begin{aligned}&\mathsf{M}\left(\dot{\mathbf{z}}, \mathbf{z}, \mathbf{u}, \mathbf{y}, \boldsymbol{\alpha}\right)\\ &=\begin{pmatrix}1\\ \dot{z}_1\end{pmatrix}\circ\ldots\circ\begin{pmatrix}1\\ \dot{z}_n\end{pmatrix}\circ\begin{pmatrix}1\\ z_1\end{pmatrix}\circ\ldots\circ\begin{pmatrix}1\\ z_n\end{pmatrix}\circ\begin{pmatrix}1\\ u_1\end{pmatrix}\circ\ldots\circ\begin{pmatrix}1\\ u_m\end{pmatrix}\\ &\ldots\circ\begin{pmatrix}1\\ y_1\end{pmatrix}\circ\ldots\circ\begin{pmatrix}1\\ y_p\end{pmatrix}\circ\begin{pmatrix}1\\ \alpha_1\end{pmatrix}\circ\ldots\circ\begin{pmatrix}1\\ \alpha_q\end{pmatrix}\in\mathbb{R}^{\overbrace{2\times\ldots\times 2}^{2n+m+p+q}},\end{aligned}$$

where the outer product provides all possible multilinear combinations of the $n \in \mathbb{N}$ state variables $\mathbf{z} \in \mathbb{R}^n$, $m \in \mathbb{N}$ input variables $\mathbf{u} \in \mathbb{R}^m$, $p \in \mathbb{N}$ output variables $\mathbf{y} \in \mathbb{R}^p$, and $q \in \mathbb{N}$ algebraic variables $\boldsymbol{\alpha} \in \mathbb{R}^q$. The monomial tensor is structurally of rank 1, i.e., there is only one outer product and not a sum of two or more necessary for representation [10].

Example 2.1: Consider an iMTI model with $n = 1$ state, and $q = 1$ algebraic variable. Thus, there are $2^3 = 8$ combinations for multilinear monomials, and the resulting monomial tensor is depicted in Fig. 2.

The total number of relevant signals of the model is $N_v = 2n + m + p + q$. In the following the notation for

$$\mathsf{M}(\dot{z}, z, \alpha) = \quad = \quad \in \mathbb{R}^{2\times 2\times 2}$$

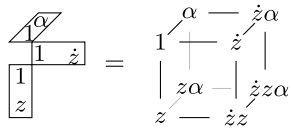


FIGURE 2. Monomial tensor of the Example 2.1.

tensor spaces is used [7]

$$\mathbb{R}^{\times^{N_v}2} := \mathbb{R}^{\overbrace{2\times\ldots\times 2}^{N_v}}.$$

The multilinear model equations are obtained from their tensors using the contracted product.

Definition 2.2: [7], [10] The *contracted product* $\left\langle \cdot \mid \cdot \right\rangle$ of a tensor $\mathsf{F} \in \mathbb{R}^{I_1\times\cdots\times I_k\times J_1\times\cdots\times J_l}$, and a tensor $\mathsf{G} \in \mathbb{R}^{I_1\times\cdots\times I_k}$, gives a tensor $\mathsf{H} = \left\langle \mathsf{F} \,\middle|\, \mathsf{G} \right\rangle \in \mathbb{R}^{J_1\times\cdots\times J_l}$ with elements

$$h_{j_1,\ldots,j_l} = \sum_{i_1=1}^{I_1}\sum_{i_2=1}^{I_2}\cdots\sum_{i_k=1}^{I_k} g_{i_1,\ldots,i_k}\, f_{i_1,\ldots,i_k,j_1,\ldots,j_l}, \tag{2}$$

where the elements are indexed by $j_l \in \{1, 2, \ldots, J_l\}$ for $l = 1, \ldots, J_l$.

With the monomial tensor and the contracted product, an iMTI model can be defined.

Definition 2.3: [7] An implicit multilinear time-invariant state-space model is defined as

$$\mathbf{0} = \left\langle \mathsf{H} \,\middle|\, \mathsf{M}\left(\dot{\mathbf{z}}, \mathbf{z}, \mathbf{u}, \mathbf{y}, \boldsymbol{\alpha}\right) \right\rangle \in \mathbb{R}^{n+Q+p}, \tag{3}$$

where the parameter tensor is $\mathsf{H} \in \mathbb{R}^{\times^{N_v}2\times(n+Q+p)}$, since there are n-state equations of the model that are constrained by $Q \in \mathbb{N}$ algebraic equations, and it has p output equations. The total number of equations is $N_\phi = n + Q + p$. The right-hand side of the iMTI model is described by the implicit multilinear function

$$\mathbf{0} = \mathbf{h}(\dot{\mathbf{z}}, \mathbf{z}, \mathbf{u}, \mathbf{y}, \boldsymbol{\alpha}), \tag{4}$$

where $\mathbf{h} : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \times \mathbb{R}^q \to \mathbb{R}^{N_\phi}$.

The full parameter tensor H of an iMTI model suffers from the curse of dimensionality since the number of elements are $N_\phi 2^{N_v}$. To overcome the curse of dimensionality, the tensor is decomposed using a normalized format of CP tensor decomposition [10], called CPN1 format [11]. The idea of the CP decomposition is to represent a tensor F by a finite sum of rank-one tensors, where the rank-one tensors are obtained by the outer product as shown in Example 2.1 [10], [21]. In [21], the *rank* of the tensor F is defined as the smallest number of rank-one tensors required to obtain an exact representation of F [10]. Determining the rank of a tensor F is an NP-hard problem [10], [22]. However, there are algorithms that can find low-rank approximations that fulfill certain accuracy requirements [10]. This work only considers exact representations with presumably non-minimal rank.

In the context of multilinear models [6], this means the representation is not necessarily minimal. The CPN1 format allows for another reduction of the number of elements to the factors by almost half [11], [23]. Using the CPN1 format, the number of elements of the parameter tensor scale linearly with number of factors $R \in \mathbb{N}$ and the number of dimensions, i.e., it has $N_\phi R$ elements, in contrast to the exponential increase $N_\phi 2^{N_v}$ of the full parameter tensor H of the iMTI model.

Definition 2.4: [5], [14] The contracted product of an iMTI model in the CPN1 format is written as a simple sum of factored multilinear polynomials

$$\mathbf{0} = \Phi\, \mathbf{s}(\mathbf{v}(t)), \tag{5a}$$

$$\mathbf{s}_r(\mathbf{v}(t)) = \prod_{i=1}^{N_v} \left(1 - |\mathbf{S}_{ir}| + \mathbf{S}_{ir}\mathbf{v}_i(t)\right), \tag{5b}$$

where $\mathbf{v}(t) = \left(\dot{\mathbf{z}}(t)^{\mathrm{T}}, \mathbf{z}(t)^{\mathrm{T}}, \mathbf{u}(t)^{\mathrm{T}}, \mathbf{y}(t)^{\mathrm{T}}, \boldsymbol{\alpha}(t)^{\mathrm{T}}\right)^{\mathrm{T}}$ is a column vector containing all states, inputs, outputs, and algebraic variables. The dependence on time is explicitly stated here to highlight that the structured contracted product at time t is a vector-valued function $\mathbf{s} : \mathbb{R}^{N_v} \to \mathbb{R}^R$, where $R \in \mathbb{N}$ is the number of factors of the parameter tensor of the iMTI model. In contrast, the time-invariant parameter matrix $\Phi \in \mathbb{R}^{N_\phi \times R}$ gives the coefficients of the iMTI model. The structural matrix $\mathbf{S} \in \mathbb{R}^{N_v \times R}$ provides the factors for the multilinear combinations of the variables in $\mathbf{v}$.

To illustrate the usage of the CPN1 format for multilinear models, one example is provided.

Example 2.2: The calculation of the active power p is a multilinear function (1), and can be written implicitly as

$$0 = p - v_a i_a - v_b i_b - v_c i_c. \tag{6}$$

It has six inputs $\mathbf{u} = \left(v_a,\ v_b,\ v_c,\ i_a,\ i_b,\ i_c\right)^{\mathrm{T}} \in \mathbb{R}^6$, and one output $y = p \in \mathbb{R}$. The parameter matrix Φ provides the coefficients of the 4 terms of (6), in this case leading to $R = 4$, thus

$$\Phi = \left(1,\ -1,\ -1,\ -1\right),$$

where each column corresponds to the combination of the variables provided by the structural matrix $\mathbf{S}$. In the structural matrix, the rows correspond to each variable, in this case the elements of $\left(\mathbf{u}^{\mathrm{T}}, y\right)^{\mathrm{T}}$ in this order, and provide the multilinear combinations of the variables used for the R-terms, such that $\mathbf{S} \in \mathbb{R}^{7\times 4}$. The second term $-V_a I_a$ of the implicit expression of (6) for instance, is indicated by the two nonzero entries $\mathbf{S}_{12} = \mathbf{S}_{42} = 1$, corresponding to $\Phi_{12} = -1$.

Following this procedure, the structural matrix of (6) is

$$\mathbf{S} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{v}(t) = \begin{pmatrix} v_a \\ v_b \\ v_c \\ i_a \\ i_b \\ i_c \\ p \end{pmatrix}.$$

Note that this is an introductory example to familiarize the reader with the notation, where $\mathbf{S}$ consists of only 1 and 0, though it could be any real number between 0 and 1. The advantage of the factorized CPN1 format becomes more evident, when considering larger examples, where good factorizations are available, see [23] or [5] for more examples. The composition of iMTI models is straightforward [7] and demonstrated by an example.

Example 2.3: Connecting two models in series is done by simply adding an algebraic constraint, for example, $0 = y_1 - u_2$, where the output of the first model y_1 is the input of the second model u_2.

The developed approaches presented in this paper use the MTI-Toolbox [24] that is available for Matlab [20]. Several examples are given in the MTI-Toolbox that demonstrate its usage, and the work [15] provides an example of a nine bus power system.

III. SYMBOLIC MULTILINEARIZATION OF CONVERTER-RELEVANT NON-MULTILINEAR DYNAMICS

There are different possibilities to obtain a multilinear model from a nonlinear power system model. Numerical multilinearization [9] generates an approximative multilinear model by, e.g., sparse gridding of the nonlinear model. This paper concerns symbolic multilinearization, by which an exact multilinear representation of the nonlinear power system model is found with the help of auxiliary variables. Some of them are already multilinear, as demonstrated in the Section I.

A. NON-MULTILINEARITIES OF POWER CONVERTERS

The controllers of power converters have several common nonlinear functions. For example, there are saturations in PI controllers, Euclidean norms for voltage or current magnitude calculations using dq-components in voltage controllers or for current saturation algorithms, trigonometric functions for Park transformations or using rotational matrices, or switching between modes of operations during, e.g., faults, [18].

The nonlinear behavior of power electronics are disregarded and ideal averaged-switching models assumed as the focus is on eigenvalue analysis. Other components in power networks are often approximated as linear models using lumped parameters, see for example [25], such as a transformer as an rl-section, or the cables as π-models.

Further examples for implicit formulations of power system components can be found in [26], where a synchronous

machine model and several controllers are written in a semi-implicit formulation, which can be easily transferred into an iMTI model. A recent report [27] shows several examples, including a synchronous generator and converters.

Next, different examples are given for the symbolic multilinearization.

B. HIGHER-ORDER POLYNOMIAL MODELS

Polynomial models can be represented by iMTI models by introducing additional algebraic variables [7]. The procedure is demonstrated by an example.

Example 3.1: Consider a first-order autonomous polynomial model

$$\dot{x} = x^2.$$

Introducing an auxiliary algebraic variable α, the polynomial model can be written as an iMTI model

$$0 = \dot{x} - x\alpha,$$
$$0 = x - \alpha.$$

C. TRIGONOMETRIC FUNCTIONS

The work [28] showed canonical expressions of different nonlinear functions by a change of variables into a polynomial formulation. This change of variables was used for modeling synchronous generators in [29], [30] to obtain a sum-of-squares formulation of the model. The representation of the nonlinear model into a polynomial model is achieved by a nonlinear change of coordinates defined by smooth mapping

$$z_1 = \cos(x), \tag{7a}$$
$$z_2 = \sin(x), \tag{7b}$$

with the algebraic constraint

$$g(\mathbf{z}) = z_1^2 + z_2^2 - 1 = 0. \tag{8}$$

Despite the change of the state-space dimensions, the work [30] showed, that for their chosen invertible transformation, their example is a Lie-Bläcklund isomorphism [31], which means that the two models have the same trajectory, and they can be called *equivalent*. The definition of a Lie-Bläcklund transformation is provided in the Appendix A-A.

The process of the change of coordinates is demonstrated using a phase-locked loop (PLL) as an example.

Example 3.2 (PLL): To simplify the explanation, the network is assumed to be in *dq*-coordinates. The PLL is depicted in Fig. 3, and the Park transformation is replaced by the rotational matrix $\mathbf{R}(\delta) : \mathbb{R}^2 \to \mathbb{R}^2$

$$\mathbf{R}(\delta) = \begin{pmatrix} \cos(\delta) & -\sin(\delta) \\ \sin(\delta) & \cos(\delta) \end{pmatrix}, \tag{9}$$

where $\delta \in \mathbb{R}$ is the angle difference between the local converter synchronous frame, denoted by *dq*, and the global rotating synchronous frame of the network denoted by *DQ* as in the work [32].

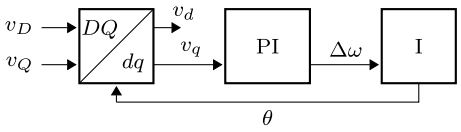


FIGURE 3. Phase-locked loop.

The PLL written in *dq*-coordinates is described by a second-order nonlinear model

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) = \begin{pmatrix} x_2 - k_\mathrm{p}\,(u_1 \sin(x_1) + u_2 \cos(x_1)) \\ -k_\mathrm{i}\,(u_1 \sin(x_1) + u_2 \cos(x_1)) \end{pmatrix} \tag{10}$$

where the states are the angle difference, and the integrator state, i.e., $\mathbf{x} = (\delta, x_\mathrm{I})^T$, and the inputs are the *DQ* voltage components, i.e., $\mathbf{u} = (V_D, V_Q)$. The gains of the PI controller are $k_\mathrm{p}, k_\mathrm{i} \in \mathbb{R}_{\geq 0}$. The right-hand side of (10) is described by the vector field $\mathbf{f} : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2$.

To transform the nonlinear model (10) into an iMTI model, the following smooth mapping is used

$$\boldsymbol{\sigma} : \begin{cases} \mathbb{R}^2 \times \mathbb{R}^2 & \to \mathcal{M} := \left\{ \mathbf{z} \in \mathbb{R}^3, \mathbf{u} \in \mathbb{R}^2 | \mathbf{g}(\mathbf{z}) = 0 \right\}, \\ (x_1, x_2, u_1, u_2) & \mapsto (\cos(x_1), \sin(x_1), x_3, u_1, u_2), \end{cases} \tag{11}$$

where (8) constraints $\mathcal{M}$. The new variables are[1]

$$\mathbf{z}(t) = \left(\cos(x_1(t)),\ \sin(x_1(t)),\ x_2(t) \right)^\mathrm{T}, \tag{12}$$

where for ease of notation the variables of the nonlinear model are set into $\boldsymbol{\kappa}(t) = (\mathbf{x}(t)^\mathrm{T}, \mathbf{u}(t)^\mathrm{T})^\mathrm{T}$, and the new variables in $\boldsymbol{\zeta} = (\mathbf{z}^\mathrm{T}, \mathbf{u}(t)^\mathrm{T})^\mathrm{T}$. Now (12) can be written as

$$\boldsymbol{\zeta}(t) = \mathbf{p}(\boldsymbol{\kappa}(t)).$$

To get the model in the new coordinates, $\boldsymbol{\zeta}(t)$ is derivated with respect to time, where the chain rule needs to be considered

$$\frac{\mathrm{d}\boldsymbol{\zeta}(t)}{\mathrm{d}t} = \frac{\mathrm{d}\mathbf{p}(\boldsymbol{\kappa})}{\mathrm{d}\boldsymbol{\kappa}(t)} \frac{\mathrm{d}\boldsymbol{\kappa}(t)}{\mathrm{d}t} = \begin{pmatrix} -\sin(x_1(t)) & 0 & 0 & 0 \\ \cos(x_1(t)) & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ 0 \\ 0 \end{pmatrix},$$

which are expressed by the new variable yielding the polynomial model

$$\dot{z}_1 = -z_2 \left(z_3 - k_\mathrm{p}\,(u_1 z_2 + u_2 z_1) \right), \tag{13a}$$
$$\dot{z}_2 = z_1 \left(z_3 - k_\mathrm{p}\,(u_1 z_2 + u_2 z_1) \right), \tag{13b}$$
$$\dot{z}_3 = -k_\mathrm{i}\,(u_1 z_2 + u_2 z_1). \tag{13c}$$

[1]The time dependence is shown here to help the reader.

Two additional algebraic variables are introduced giving the iMTI model of the PLL

$$\mathbf{0} = \begin{pmatrix} \dot{z}_1 + \left(z_3 - k_\mathrm{p}\left(u_1 z_2 + u_2 z_1\right)\right)\alpha_2 \\ \dot{z}_2 - \left(z_3 - k_\mathrm{p}\left(u_1 z_2 + u_2 z_1\right)\right)\alpha_1 \\ \dot{z}_3 + k_\mathrm{i}\left(u_1 z_2 + u_2 z_1\right) \\ z_1 - \alpha_1 \\ z_2 - \alpha_2 \end{pmatrix}, \tag{14}$$

where the right-hand side is the implicit multilinear function $\mathbf{h} : \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^5$.

In the Appendix A-B, it is shown that (11) is a Lie-Bläcklund transformation.

D. ON THE EXISTENCE OF A SOLUTION AND NUMERICAL TREATMENT

Contrary to the ODEs, where there always exists a unique solution for an initial value if the right-hand side is smooth, the solution of general nonlinear DAEs may not be straightforward [33]. However, since the iMTI model of the power system is derived from a nonlinear ODE model, it is assumed that there exists a unique solution for the multilinear counterpart. For the simulation of the iMTI model, the provision of the correct initial conditions that fulfill the algebraic constraints for the iMTI model is crucial to obtain a unique solution.

IV. EFFICIENT LINEARIZATION OF IMPLICIT MULTILINEAR MODELS

The multilinear formulation of the power system components allow for an efficient computation of the Jacobian [5]. Therefore, a linear model can be easily obtained, which is outlined in the following section.

A. EFFICIENT COMPUTATION OF THE JACOBIAN

The stucture of the CPN1 format as a factorized multilinear polynomial allows for an efficient computation of the Jacobian matrix. The examples in [5] showed a gain of computational performance by two orders of magnitude. This paper extends the method to iMTI models by calculating the partial derivates of (3) with respect to all variables contained in the signal vector $\mathbf{v}$. Taking the partial derivative with respect to the i-th variable of the iMTI model (5), means computing the partial derivative of the structured contracted product (5b)

$$\left.\frac{\partial \mathbf{s}_r(\mathbf{v})}{\partial \mathbf{v}_i}\right|_{\mathbf{v}=\bar{\mathbf{v}}} = \mathbf{S}_{ir} \prod_{\substack{j=1 \\ j\neq i}}^{N_\mathrm{v}} \left(1 - |\mathbf{S}_{jr}| + \mathbf{S}_{jr}\bar{\mathbf{v}}_j\right). \tag{15}$$

From (15), it can be seen that i-th term in $\mathbf{s}(\bar{\mathbf{v}})$ needs to be canceled out and the factor $\mathbf{S}_{ir}$ remains. This structure is used to construct a numerically efficient procedure to calculate the Jacobian, which also works for sparse structrural matrices. Firstly, by computing a matrix $\mathbf{D} \in \mathbb{R}^{R \times N_\mathrm{v}}$ with elements

$$\mathbf{D}_{ir} = \begin{cases} \dfrac{\mathbf{S}_{ir}}{1 - |\mathbf{S}_{ir}| + \mathbf{S}_{ir}\bar{\mathbf{v}}_i} & \text{iff } \mathbf{s}_r(\bar{\mathbf{v}}) \neq 0, \\ & \quad \forall\, r = 1, \ldots, R, \\ \mathbf{S}_{ir} & \text{iff } \left(1 - |\mathbf{S}_{ir}| + \mathbf{S}_{ir}\bar{\mathbf{v}}_i\right) = 0, \\ 0 & \text{else,} \end{cases}$$

and secondly, by evaluating the structured contracted at the operating point $\bar{\mathbf{v}}$

$$\mathbf{s}_r(\bar{\mathbf{v}}) = \begin{cases} \displaystyle\prod_{j=1}^{N_\mathrm{v}} \left(1 - |\mathbf{S}_{jr}| + \mathbf{S}_{jr}\bar{\mathbf{v}}_j\right) & \text{iff } \mathbf{s}_r(\bar{\mathbf{v}}) \neq 0, \\ & \quad \forall\, r = 1, \ldots, R, \\ \displaystyle\prod_{\substack{j=1 \\ j\neq i}}^{N_\mathrm{v}} \left(1 - |\mathbf{S}_{jr}| + \mathbf{S}_{jr}\bar{\mathbf{v}}_j\right) & \text{iff} \\ & \left(1 - |\mathbf{S}_{ir}| + \mathbf{S}_{ir}\bar{\mathbf{v}}_i\right) = 0, \\ 0 & \text{else.} \end{cases}$$

Then the calculation of the Jacobian (15) becomes a simple matrix multiplication

$$\mathbf{J} = \Phi(\mathbf{D} \boxdot \mathbf{s}(\bar{\mathbf{v}}))^\mathrm{T} \in \mathbb{R}^{N_\phi \times N_\mathrm{v}}, \tag{16}$$

where the *broadcast product* [34], often used in scientific computing, e.g., Matlab, is denoted by $\boxdot$ [35]. In this case, the elements of $\mathbf{s}(\bar{\mathbf{v}})$ are duplicated to match the dimensions of $\mathbf{D}$ allowing for an element-wise multiplication, i.e., the Hadamard product. More aspects for the implementation can be found in [5], which are similar to the case of eMTI models.

B. LINEAR DESCRIPTOR SYSTEMS

Definition 4.1: A continuous-time LDSS is described by [36]

$$\mathbf{E}\begin{pmatrix} \dot{\mathbf{z}} \\ \dot{\boldsymbol{\alpha}} \end{pmatrix} = \mathbf{A}\begin{pmatrix} \mathbf{z} \\ \boldsymbol{\alpha} \end{pmatrix} + \mathbf{B}\mathbf{u}, \tag{17}$$

where $\mathbf{z} \in \mathbb{R}^n$, and $\boldsymbol{\alpha} \in \mathbb{R}^q$, from the multilinear model, form the state vector of the LDSS model. The system matrix is denoted by $\mathbf{A} \in \mathbb{R}^{(n+q)\times(n+q)}$, with the descriptor matrix $\mathbf{E} \in \mathbb{R}^{(n+q)\times(n+q)}$ having the same dimensions, and $m \in \mathbb{N}$ inputs of the system are given by the input matrix $\mathbf{B} \in \mathbb{R}^{(n+q)\times m}$.

To obtain an LDSS (17) from an iMTI model, at the equilibrium point, meaning $\dot{\mathbf{z}} = \mathbf{0}$, and $\dot{\boldsymbol{\alpha}} = \mathbf{0}$, the matrices are extracted from the Jacobian (16)

$$\begin{aligned} \mathbf{E} &= -\left.\begin{pmatrix} \frac{\partial \mathbf{h}(\mathbf{v})}{\partial \dot{\mathbf{z}}} & \frac{\partial \mathbf{h}(\mathbf{v})}{\partial \dot{\boldsymbol{\alpha}}} \end{pmatrix}\right|_{\mathbf{v}=\bar{\mathbf{v}}}, \\ \mathbf{A} &= \left.\begin{pmatrix} \frac{\partial \mathbf{h}(\mathbf{v})}{\partial \mathbf{z}} & \frac{\partial \mathbf{h}(\mathbf{v})}{\partial \boldsymbol{\alpha}} \end{pmatrix}\right|_{\mathbf{v}=\bar{\mathbf{v}}}, \quad \mathbf{B} = \left.\begin{pmatrix} \frac{\partial \mathbf{h}(\mathbf{v})}{\partial \mathbf{u}} \end{pmatrix}\right|_{\mathbf{v}=\bar{\mathbf{v}}}, \end{aligned} \tag{18}$$

where all variables are in $\mathbf{v} = \left(\dot{\mathbf{z}}^\mathrm{T}, \dot{\boldsymbol{\alpha}}^\mathrm{T}, \mathbf{z}^\mathrm{T}, \boldsymbol{\alpha}^\mathrm{T}, \mathbf{u}^\mathrm{T}\right)^\mathrm{T}$. The minus sign results from the implicit notation in (4) and rearranging the descriptor matrix $\mathbf{E}$ to the left-hand side.

With the obtained matrices, the linearized iMTI model can be written as an LDSS. Note that the coordinate system is moved into the equilibrium point, i.e.,

$$\mathbf{E}\begin{pmatrix}\dot{\mathbf{z}}\\ \dot{\boldsymbol{\alpha}}\end{pmatrix} = \mathbf{A}\begin{pmatrix}\mathbf{z}-\bar{\mathbf{z}}\\ \boldsymbol{\alpha}-\bar{\boldsymbol{\alpha}}\end{pmatrix} + \mathbf{B}\left(\mathbf{u}-\bar{\mathbf{u}}\right),$$

for instance for (17).

If $\det(\mathbf{E}) \neq 0$, it can be inverted and (17) can be written explicitly as

$$\begin{pmatrix}\dot{\mathbf{z}}\\ \dot{\boldsymbol{\alpha}}\end{pmatrix} = \mathbf{E}^{-1}\mathbf{A}\begin{pmatrix}\Delta\mathbf{z}\\ \Delta\boldsymbol{\alpha}\end{pmatrix} + \mathbf{E}^{-1}\mathbf{B}\mathbf{u}. \tag{19}$$

If the iMTI model has a low number of factors, then it can be advantageous to store the LDSS model in the CPN1 format. Furthermore, affine models that occur when the state derivatives are unequal to zero and a Taylor approximation is used, can also be modeled in the CPN1 format.

Example 4.1 (PLL - continued): Linearizing the iMTI model of the PLL by the computation of the Jacobian, and extracting the matrices using (18), yields the following linear descriptor model

$$\mathbf{E} = \begin{pmatrix}-1 & 0 & 0 & 0 & 0\\ 0 & -1 & 0 & 0 & 0\\ 0 & 0 & -1 & 0 & 0\\ 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0\end{pmatrix},$$

$$\mathbf{A} = \begin{pmatrix}-k_\mathrm{p}\alpha_2 u_2 & -k_\mathrm{p}\alpha_2 u_1 & \alpha_2 & 0 & z_3 - g(\mathbf{z})\\ k_\mathrm{p}\alpha_1 u_2 & k_\mathrm{p}\alpha_1 u_1 & -\alpha_1 & g(\mathbf{z}) - z_3 & 0\\ k_\mathrm{I} u_2 & k_\mathrm{I} u_1 & 0 & 0 & 0\\ 1 & 0 & 0 & -1 & 0\\ 0 & 1 & 0 & 0 & -1\end{pmatrix}\Bigg|_{\substack{\mathbf{z}=\bar{\mathbf{z}},\\ \boldsymbol{\alpha}=\bar{\boldsymbol{\alpha}},\\ \mathbf{u}=\bar{\mathbf{u}}}},$$

where $g(\mathbf{z}) = k_\mathrm{p}\left(z_1 u_2 + z_2 u_1\right)$, and

$$\mathbf{B} = \begin{pmatrix}-k_\mathrm{P} z_2\,\alpha_2 & -k_\mathrm{P}\, z_1\,\alpha_2\\ k_\mathrm{P}\, z_2\,\alpha_1 & k_\mathrm{P}\, z_1\,\alpha_1\\ k_\mathrm{I}\, z_2 & k_\mathrm{I}\, z_1\\ 0 & 0\\ 0 & 0\end{pmatrix}\Bigg|_{\substack{\mathbf{z}=\bar{\mathbf{z}},\\ \boldsymbol{\alpha}=\bar{\boldsymbol{\alpha}},\\ \mathbf{u}=\bar{\mathbf{u}}}}.$$

V. SMALL-SIGNAL STABILITY ANALYSIS BY GENERALIZED EIGENVALUES

The stability of operating points of nonlinear power systems in the sense of Lyapunov can be assessed locally by checking the eigenvalues of the linearized model [19, p. 49]. Often these are linear eigenvalue problems, while for linear descriptor systems, the generalized eigenvalue problem applies [37]. Therefore, the stability of the operating points of iMTI models can be assessed by analyzing the generalized eigenvalue of the obtained LDSS, which is described in this section.

The generalized eigenvalue problem (GEP) is [37]

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{E}\mathbf{v}, \tag{20a}$$
$$\mathbf{w}\mathbf{A} = \lambda\mathbf{w}\mathbf{E}, \tag{20b}$$

where $\lambda \in \mathbb{C}$ is the eigenvalue, $\mathbf{v} \in \mathbb{C}^{(n+q)\times 1}$ the corresponding right eigenvector, and $\mathbf{w} \in \mathbb{C}^{1\times(n+q)}$ the left eigenvector. If $\mathbf{E}$ is equal to the identity matrix, meaning there are no algebraic equations, the GEP reduces to the common eigenvalue problem. The matrix pair $(\mathbf{E}, \mathbf{A})$ form a matrix pencil. A matrix pencil is a matrix-valued polynomial function, though for LDSS, it is a linear matrix pencil $\lambda\mathbf{E}-\mathbf{A}$, which is a family of matrices parameterized by a complex number $\lambda \in \mathbb{C}$ [19, p. 66] [38]. Due to the construction of the LDSS, i.e., $\mathbf{E}$ and $\mathbf{A}$ are square matrices, the matrix pencil of the derived LDSS from the iMTI model has a characteristic polynomial. The generalized eigenvalues are the roots of the characteristic polynomial of the matrix pencil $(\mathbf{A}, \mathbf{E})$. The order of the characteristic polynomial are the k finite eigenvalues of the regular pencil. The regular pencil can have l infinite eigenvalues, where $k + l = n + q$. Considering the reciprocal generalized eigenvalue problem $\lambda^{-1}\mathbf{A}\mathbf{v} = \mathbf{E}\mathbf{v}$, if $\mathbf{E}$ is singular, there exists a null vector $\mathbf{v}$, such that $\mathbf{E}\mathbf{v} = \mathbf{0} \in \mathbb{R}^{(n+q)\times 1}$. Thus, $\lambda^{-1}\mathbf{A}\mathbf{v} = \mathbf{0} \in \mathbb{R}^{(n+q)\times 1}$, so that $\lambda^{-1} = 0$, corresponing to $\lambda \to \infty$ [37]. This case is demonstrated in the following.

Example 5.1: Consider the following autonomous descriptor model with two states

$$\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}\begin{pmatrix}\dot{z}_1\\ \dot{z}_2\end{pmatrix} = \begin{pmatrix}0 & 1\\ 1 & 1\end{pmatrix}\begin{pmatrix}z_1\\ z_2\end{pmatrix}.$$

The descriptor matrix $\mathbf{E}$ is singular since $\det(\mathbf{E}) = 0$, meaning there is one right eigenvector that is in the nullspace of $\mathbf{E}$.

Computing the generalized eigenvalues and eigenvectors using Matlab's `eig` command yields $\lambda_1 = -1$, and $\lambda_2 = \texttt{Inf}$, and

$$\mathbf{V} = \begin{pmatrix}1 & 0\\ -1 & 1\end{pmatrix},\ \mathbf{W} = \begin{pmatrix}1 & 0\\ -1 & 1\end{pmatrix},$$

where the second right eigenvector gives the zero vector

$$\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}\begin{pmatrix}0\\ 1\end{pmatrix} = \begin{pmatrix}0\\ 0\end{pmatrix},$$

corresponding to the infinite eigenvalue, i.e., the null space of $\mathbf{E}$.

Note that in this example, the LDSS can be easily transformed into an LSS by recognizing that the second equation, $z_2 = -z_1$, reveals the eigenvalue $\lambda = -1$.
The infinite generalized eigenvalues are a result of the algebraic equations, which can be interpreted as infinitely fast [26] and thus are not of interest for the small-signal stability analysis. For further discussion on the participation of algebraic variable in the modes, see [39].

The generalized eigenvalues are used to assess if an equilibrium point of the iMTI model is unstable, according to the following definition.

Definition 5.1: [19] The equilibrium point ($\bar{\mathbf{v}}$) of the iMTI model is locally unstable if at least one of the generalized eigenvalues of the LDSS are on the right-half of the complex plane, i.e., $\Re(\lambda) > 0$.
With this definition, the small-signal stability of an equilibrium point of the PLL is assessed in the following using the generalized eigenvalues.

Example 5.2 (PLL - continued): The generalized eigenvalues of the obtained linear descriptor model of the PLL given in Example 4.1 are evaluated in the following. The rank of the descriptor matrix $\text{rank}(\mathbf{E}) = 3$, therefore, two generalized eigenvalues will be infinite. The system matrix is of $\text{rank}(\mathbf{A}) = 4$, thus there will be one zero eigenvalue. The PLL has the proportional and integral gains $k_\text{P} = 0.5\,\text{rad}\,\text{V}^{-1}\,\text{s}^{-1}$, and $k_\text{I} = 9\,\text{rad/V/s}^2$, respectively. After an angle step of $\delta = 4°$, the iMTI model reaches the new equilibrium

$$\bar{z}_1 = \bar{z}_4 = 0.9976, \bar{z}_2 = \bar{z}_5 = -0.0698, \quad \bar{z}_3 = 0\,\text{rad}\,\text{s}^{-1},$$
$$\bar{u}_1 = 325.2059\,\text{V}, \bar{u}_2 = 22.7406\,\text{V}.$$

At this equilibrium point, the LDSS has the generalized eigenvalues

$$\lambda_1 = -20.6046, \quad \lambda_2 = -142.3954, \quad \lambda_3 = 0, \quad \lambda_{4,5} = \texttt{Inf},$$

which means that the equilibrium point is stable.

For a comparison, the Jacobian of the nonlinear model of the PLL (10) is computed with respect to x_1, x_2

$$\mathbf{A} = \begin{pmatrix} -k_\text{P}\,(u_1 \cos(x_1) - u_2 \sin(x_1)) & 1 \\ -k_\text{I}\,(u_1 \cos(x_1) - u_2 \sin(x_1)) & 0 \end{pmatrix},$$

which yields the eigenvalues

$$\lambda_1 = -142.3954, \quad \lambda_2 = -20.6046.$$

Therefore, both models come to the same conclusion that the studied equilibrium point is stable.

VI. POWER SYSTEM MODEL

To test the proposed methodology, an essential power system network that reflects the relevant dynamics of converter-dominated networks is constructed. It consists of a grid-forming (GFM) converter, a grid-following (GFL) converter, and a load, all connected to a Thévenin-equivalent grid as depicted in Fig. 4. This network configuration allows triggering a number of disturbances typically encountered in power systems such as changes in load, voltage amplitude, phase angle or frequency. Therefore, the network is complex enough to conduct typical scenarios in power system simulation studies as well as small-signal stability analysis. The power system model is constructed in the following section.

The models are implemented in Matlab [20] using the MTI-Toolbox [24], as well as in Simulink for a comparison.

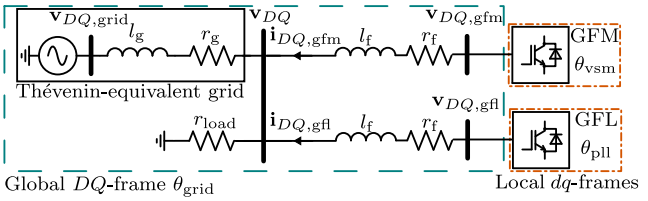


FIGURE 4. Scheme of the essential system in the global frame θ_grid (dashed line), and the local frames of the converters (dashed-dotted line).

A. SYSTEM DESCRIPTION IN DQ-COORDINATES

Similar to other small-signal power system stability studies [32], [40], the network is modeled in dq-coordinates. Each device has its own rotating dq-frame, indicated by the lower case, while the network is modeled in a global rotating frame indicated by the upper case DQ. This modeling approach helps in the linearization of the model because for small angle deviations between the global and the local frame, the trigonometric functions in the rotational matrices can be linearly approximated [32]. Therefore, the rotational matrix $\mathbf{R}(\theta_i, \theta_\text{grid}) \in \mathbb{R}^{2\times 2}$ is applied to the grid voltage and current, to rotate from $DQ \to dq$

$$\mathbf{R}(\theta_i, \theta_\text{grid}) = \begin{pmatrix} h_1(\theta_i, \theta_\text{grid}) & h_2(\theta_i, \theta_\text{grid}) \\ -h_2(\theta_i, \theta_\text{grid}) & h_1(\theta_i, \theta_\text{grid}) \end{pmatrix}$$
$$\begin{aligned} h_1(\theta_i, \theta_\text{grid}) &= \cos(\theta_i - \theta_\text{grid}) \\ &= \cos(\theta_i)\cos(\theta_\text{grid}) + \sin(\theta_i)\sin(\theta_\text{grid}) \\ h_2(\theta_i, \theta_\text{grid}) &= \sin(\theta_i - \theta_\text{grid}) \\ &= \sin(\theta_i)\cos(\theta_\text{grid}) - \cos(\theta_i)\sin(\theta_\text{grid}), \end{aligned} \quad (21)$$

where $\theta_i \in \mathbb{R}$ is the phase angle of the local device, and $\theta_\text{grid} \in \mathbb{R}$ is the angle of the global synchronous reference frame (SRF). The inverse rotation $\mathbf{R}^{-1}(\theta_i, \theta_\text{grid})$ is used to rotate from $dq \to DQ$.

B. NETWORK MODEL

The network operates at high voltage 230 kV with 50 Hz. The grid connection is modeled as a Thévenin equivalent with a short circuit ratio (SCR) of 5, a base apparent power of 100 MVA, and an X/R-ratio of 10.

For a general rl circuit between two nodes k and l, with voltages $\mathbf{v}_{DQ,k}, \mathbf{v}_{DQ,l} \in \mathbb{R}^2_{\geq 0}$, and with the current $\mathbf{i}_{DQ,kl} \in \mathbb{R}^2$ flowing from k to l, the current is described by

$$\begin{aligned} \dot{\mathbf{i}}_{DQ,kl} = &\left(-\frac{r_{kl}}{l_{kl}} - \omega_g \mathcal{J}_2\right)\mathbf{i}_{DQ,kl} \\ &+ \frac{1}{l_{kl}}\left(\mathbf{v}_{DQ,k} - \mathbf{v}_{DQ,l}\right), \end{aligned} \quad (22)$$

where $r_{kl}, l_{kl} \in \mathbb{R}_{>0}$ are the resistance and inductance values of the section kl, the angular grid frequency $\omega_\text{g} \in \mathbb{R}_{>0}$, and for the cross-coupling in dq-coordinates $\mathcal{J}_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. If the grid frequency is either a state or an output of the system, the rl-model is multilinear. The rl circuit model (22)

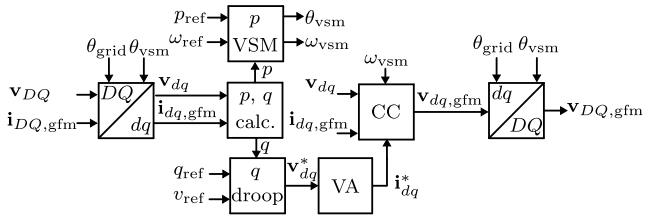


FIGURE 5. Cascaded control loop structure of the grid-forming converter.

is used for the Thévenin equivalent grid, and for the filter equations of the converters.

The network has a resistive load $r_{\text{load}} \in \mathbb{R}_{>0}$ which yields the voltage

$$\mathbf{v}_{DQ} = r_{\text{load}} \Delta \mathbf{i}_{DQ}, \tag{23}$$

where $\Delta \mathbf{i}_{DQ}$ is the difference between the currents flowing into the node.

C. GRID-FORMING CONVERTER

A virtual synchronous machine (VSM) [41] is used for the active power control of the GFM converter, and a filtered voltage droop control for the reactive power control [40]. It was shown that different GFM control strategies behave like a droop-based controller in a small-signal regime [40], [42]. The implemented GFM controller uses a virtual admittance (VA) cascaded by a current control (CC)-loop, as depicted in Fig. 5.

After rotating the voltage $\mathbf{v}_{DQ} \in \mathbb{R}^2$, and the current $\mathbf{i}_{DQ} \in \mathbb{R}^2$, to the local coordinates using (21), i.e, $\mathbf{v}_{dq} \in \mathbb{R}^2$, and $\mathbf{i}_{dq} \in \mathbb{R}^2$, respectively, the active power $p \in \mathbb{R}$, and the reactive power $q \in \mathbb{R}$ are calculated.

1) ACTIVE POWER CONTROL

The active power control is based on the VSM approach, called synchronverter [41], that tries to mimic the frequency dynamics of a synchronous generator (SG) [41], [43]

$$\dot{\theta}_{\text{vsm}} = \omega_{\text{b}} \omega_{\text{vsm}}, \tag{24a}$$

$$2\, h \dot{\omega}_{\text{vsm}} = (p_{\text{ref}} - \frac{p}{s_{\text{b}}}) + k_{\text{d}}(\omega_{\text{vsm}} - \omega_{\text{ref}}), \tag{24b}$$

where the controller uses the p.u. quantities for the active power reference $p_{\text{ref}} \in (-1, 1)$, and the angular frequency reference $\omega_{\text{ref}} \in [0, 1)$, the base angular frequency $\omega_{\text{b}} \in \mathbb{R}_{>0}$, and the base apparent power $s_{\text{b}} \in \mathbb{R}_{>0}$. The angular frequency dynamics $\omega_{\text{VSM}} \in \mathbb{R}$ depend on the virtual inertia's constant $h \in \mathbb{R}_{\geq 0}$, that can be chosen to meet the system inertia requirement [43, Sec. V-d], and the choice of the (virtual) damping $k_{\text{d}} \in \mathbb{R}_{\geq 0}$, which is a design choice and [43, Sec. V-d] provides a guideline.

The rotational matrix (21) uses the angle θ_{VSM}. Therefore, two additional states are introduced following the procedure outlined in Section III-C

$$\dot{z}_{\cos,\text{vsm}} = -\omega_{\text{b}} \omega_{\text{vsm}} z_{\sin,\text{vsm}}, \tag{25a}$$

$$\dot{z}_{\sin,\text{vsm}} = \omega_{\text{b}} \omega_{\text{vsm}} z_{\cos,\text{vsm}}, \tag{25b}$$

where the constraint as in (8) is not explicitly given here.

2) REACTIVE POWER CONTROL

For the reactive power control, a droop control with a low-pass filter is used [40],

$$\dot{x}_{q,\text{filt}} = -\omega_f x_{q,\text{filt}} + \omega_f q, \tag{26a}$$

$$v^*_{\text{d}} - v_{\text{ref}} = -k_{\text{q}} \left(x_{q,\text{filt}} - q_{\text{ref}} \right), \tag{26b}$$

where $x_{q,\text{filt}} \in \mathbb{R}$ is the integrator state of the low-pass filter, $\omega_f \in \mathbb{R}_{\geq 0}$ is the cut-off frequency, and $k_{\text{q}} \in \mathbb{R}_{\geq 0}$ is a proportional gain that emulates the droop characteristic of the SG. The reactive power reference q_{ref} is given in var, and $v_{\text{peak}} \in \mathbb{R}_{>0}$ is the nominal peak line-to-ground voltage in V, so that the generated reference value of the d-component $v^*_{\text{d}} \in \mathbb{R}_{>0}$ is also in V. Note that $v_{\text{q,ref}} = 0$ for the voltage regulation.

3) VIRTUAL ADMITTANCE

A virtual admittance can reduce the instantaneous change in active power of the GFM converter to large disturbances by adjusting the current reference $\mathbf{i}^*_{dq} \in \mathbb{R}^2$. Lowering the instantaneous change in active power can help the converter to withstand larger disturbances, e.g., large phase jumps of 30°, and remain connected [43, Sec. IV.-D]. A state-space representation of the virtual admittance is

$$\dot{\mathbf{x}}_{\text{va},dq} = \left(-\frac{r_{\text{v}}}{l_{\text{v}}} + \omega_{\text{b}} \mathcal{J}_2 \right) \mathbf{x}_{\text{va},dq} + \mathbf{v}_{dq,\text{ref}}, \tag{27a}$$

$$\mathbf{i}^*_{dq} = \frac{1}{l_{\text{v}}} \mathbf{x}_{\text{va},dq}, \tag{27b}$$

where the virtual inductance and the virtual resistance are $l_{\text{v}}, r_{\text{v}} \in \mathbb{R}_{>0}$, respectively. The integrator state of the virtual admittance is denoted by $\mathbf{x}_{\text{va,dq}} \in \mathbb{R}^2$. The cross-coupling between the dq-axis is indicated by $\mathcal{J}_2$. According to [44], the output inductance of SG is 30 % to 50 % and is used as a reference for the range for the virtual inductor $k_{\text{va}} \in [0, 1)$. The virtual resistor should be around one third of the virtual inductor $r_{\text{v}} = k_{\text{va}} \frac{z_{\text{b}}}{3}$ with $l_{\text{v}} = k_{\text{va}} z_{\text{b}} / (\omega_{\text{b}})$ [44].

4) CURRENT CONTROL

To follow the current reference from the virtual admittance, a common current control loop using a PI control is used. It is based on the internal model principle considering the converter filter [45]. The multilinear model of the current control loop is

$$\dot{\mathbf{x}}_{i,dq} = k_{i,\text{i}} \left(\mathbf{i}_{\text{dq,ref}} - \mathbf{i}_{dq} \right), \tag{28a}$$

$$\begin{aligned} \mathbf{v}_{dq,\text{gfm}} = {} & k_{i,\text{p}} \left(\mathbf{i}_{dq,\text{ref}} - \mathbf{i}_{dq} \right) + k_{i,\text{i}} \mathbf{x}_{I,dq} \\ & + l_f \omega_{\text{b}} \omega_{\text{vsm}} \mathcal{J}_2 \mathbf{i}_{dq} + \mathbf{v}_{dq}, \end{aligned} \tag{28b}$$

where $\mathbf{x}_{i,dq} \in \mathbb{R}^2$ denotes the integrator state, the gains of the PI controller are $k_{i,\text{i}}, k_{i,\text{p}} \in \mathbb{R}_{\geq 0}$, and $\mathbf{u}_{dq,\text{ref}} \in \mathbb{R}^2$ is the reference voltage that is usually sent to the pulse-width modulation signal generator to obtain the gating signal for the switches, but here considered as the output

voltage of the converter. The inverse rotation of (21) is applied to $\mathbf{v}_{dq,\mathrm{gfm}}$ to get the output voltage of the converter in the global coordinate frame. The closed-loop response of the PI-controller is designed as a first-order response with $\tau_{\mathrm{cc}} = 1\,\mathrm{ms}$, and $k_{i,\mathrm{p}} = l_f/\tau_{\mathrm{cc}}$, and $k_{i,\mathrm{i}} = r_f/\tau_{\mathrm{cc}}$ for the gains [46].

TABLE 1. Converter parameters.

Symbol	Value	Symbol	Value
s_{b}	100 MVA	r_{f}	0.02 p.u.
l_{f}	0.07 p.u.	h	1 s
k_{d}	50	ω_{ref}	1 p.u.
ω_f	$20\,\mathrm{rad\,s^{-1}}$	k_q	$3.7559 \cdot 10^{-4}\,\mathrm{V\,var^{-1}}$
$k_{\mathrm{p,CC}}$	$117.87\,\mathrm{H\,s^{-1}}$	$k_{\mathrm{i,CC}}$	$1058\,\Omega\,\mathrm{s^{-1}}$
r_{v}	$0.30\,\Omega$	l_{v}	0.03 H
$k_{\mathrm{p,PLL}}$	$1.065 \cdot 10^{-4}\,\mathrm{rad\,V^{-1}\,s^{-1}}$	$k_{\mathrm{I,PLL}}$	$0.0011\,\mathrm{rad\,V^{-1}\,s^{-2}}$
$k_{\mathrm{p},p}$	$10 \cdot 10^{-5}\,\mathrm{A\,W^{-1}}$	$k_{\mathrm{i},p}$	$1 \cdot 10^{-2}\,\mathrm{A\,W^{-1}\,s^{-1}}$
$k_{\mathrm{p},q}$	$10 \cdot 10^{-5}\,\mathrm{A\,var^{-1}}$	$k_{\mathrm{i},q}$	$1 \cdot 10^{-2}\,\mathrm{A\,var^{-1}\,s^{-1}}$

D. GRID-FOLLOWING CONVERTER

The cascaded controller structure of the GFL converter is depicted Fig. 6. The GFL converter uses a PLL to synchronize with the grid voltage, as shown in (14), which is tuned following [47], where the gains are $k_{\mathrm{p,PLL}}$, and $k_{\mathrm{i,PLL}}$. A pq-controller is used to regulate the power exchange of the converter with the network by providing the current reference signal i^*_{dq}. The active and reactive power are controlled by two separate PI-controllers with the gains $k_{\mathrm{p},pq}$, and $k_{\mathrm{i},pq}$. The current controller then uses i^*_{dq} to generate the voltage at the converter terminal $\mathbf{v}_{dq,\mathrm{gfl}}$, and it has the same structure and tuning as in (28a).

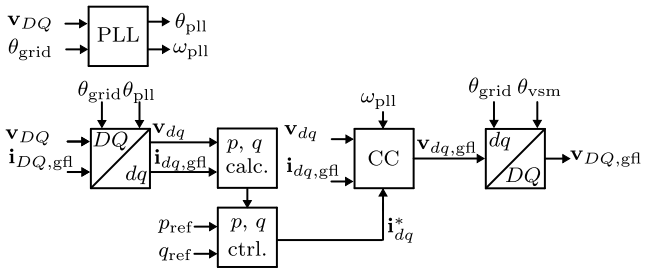


FIGURE 6. Control block diagram of the grid-following converter.

E. COMPLETE MULTILINEAR MODEL

The tuning and sizing values of the GFM and GFL converter are listed in Table 1, besides the parameter in Section VI-B. Fig. 7 depicts the steps for constructing an iMTI model using theMTI-Toolbox [24] , and this case study is available in the demo files called `demoSSA`. The model construction follows a modular approach. For each sub-model in step 1.a), for instance, the current control loop of the converter, an iMTI model is built into an object specific to the MTI-Toolbox.

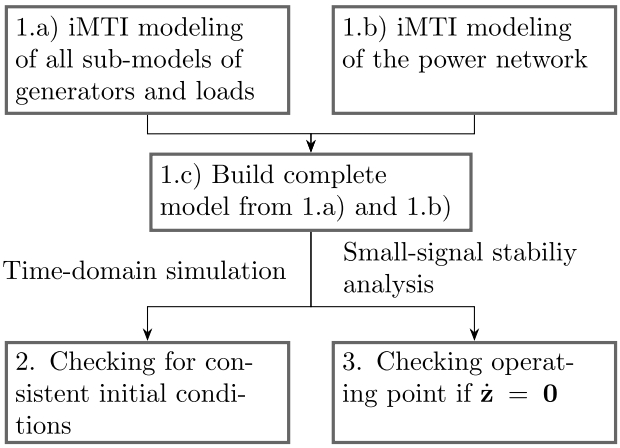


FIGURE 7. Steps for constructing an iMTI power system model.

The sub-model construction can be done either by parsing symbolic equations, from which the CPN1 decomposed tensor is automatically built, or by providing the structural matrix and the parameter matrix, as shown in Example 2.2. Each sub-model gets unique state, input, and algebraic names, which are used later to connect the sub-models and construct the model of a generator or a load. The power network structure needs to be modeled as depicted in Fig. 7 1.b), which indicates the connections of the generation or load unit. With the unique identifiers of each sub-model and the power network structure, the complete model is constructed in 1.c). Then, for time-domain simulations, there is a check for consistent initial conditions in 2., e.g., if (8) is fulfilled. Note that there are algorithms to compute consistent initial conditions. For small-signal stability analysis, in 3. it is checked if, for the chosen operating point, the model is at equilibrium. While these steps are script-based, they can be connected to a graphical interface, similar to MATLAB Simulink.

The total iMTI model has the following states of each subsystem, and the input vector for the whole model, as follows

$$\mathbf{z}_{\mathrm{gfm}} = (\omega_{\mathrm{vsm}},\ \theta_{\mathrm{VSM}},\ z_{\cos,\mathrm{vsm}},\ z_{\sin,\mathrm{vsm}},\ x_{q,\mathrm{filt}},\ \mathbf{x}_{\mathrm{va},dq},\ \mathbf{x}_{i,dq})^{\mathrm{T}} \in \mathbb{R}^9, \tag{29a}$$

$$\mathbf{z}_{\mathrm{gfl}} = (z_{\cos,\mathrm{pll}},\ z_{\sin,\mathrm{pll}},\ x_{\mathrm{i,pll}},\ \theta_{\mathrm{pll}},\ x_{\mathrm{i},p},\ x_{\mathrm{i},q},\ \mathbf{x}_{\mathrm{i},dq})^{\mathrm{T}} \in \mathbb{R}^9, \tag{29b}$$

$$\mathbf{z}_{\mathrm{grid}} = (\mathbf{i}_{DQ,\mathrm{grid}},\ \mathbf{i}_{DQ,\mathrm{gfm}},\ \mathbf{i}_{DQ,\mathrm{gfl}},\ z_{\cos,\mathrm{grid}},\ z_{\sin,\mathrm{grid}})^{\mathrm{T}} \in \mathbb{R}^8, \tag{29c}$$

$$\mathbf{u}_{\mathrm{total}} = (p_{\mathrm{ref,gfm}},\ q_{\mathrm{ref,gfm}},\ \omega_{\mathrm{ref}},\ v_{\mathrm{ref}},\ p_{\mathrm{ref,gfl}},\ q_{\mathrm{ref,gfl}})^{\mathrm{T}} \in \mathbb{R}^6, \tag{29d}$$

where $\mathbf{z}_{\mathrm{total}} = \left(\mathbf{z}^{\mathrm{T}}_{\mathrm{gfm}}, \mathbf{z}^{\mathrm{T}}_{\mathrm{gfl}}, \mathbf{z}^{\mathrm{T}}_{\mathrm{grid}}\right)^{\mathrm{T}}$. The output and intermediate variables are not explicitly stated here. The iMTI model of the total network in CPN1 has 165 factors, with 58 equations.

There is a special case for the calculation of the Jacobian because of the nonlinear change of coordinates to represent

the trigonometric functions, as in the PLL in Example 3.2. If the partial derivative of the iMTI needs to be taken with respect to a variable of the original model, which are in this example angles θ, then the chain rule applies

$$\left.\frac{\partial \mathbf{h}(\mathbf{v})}{\partial \mathbf{x}_i}\right|_{\mathbf{v}=\bar{\mathbf{v}}} = \left.\frac{\partial \mathbf{h}(\mathbf{v})}{\partial \mathbf{v}_i}\right|_{\mathbf{v}=\bar{\mathbf{v}}} \left.\frac{\partial \mathbf{f}(\kappa)}{\partial \kappa_i}\right|_{\kappa=\bar{\kappa}},$$

which generalizes to

$$\mathbf{J}_{\sigma\circ\mathbf{f}}(\bar{\mathbf{x}}, \bar{\mathbf{u}}) = \mathbf{J}_{\sigma}(\mathbf{f}(\bar{\mathbf{x}}, \bar{\mathbf{u}}))\, \mathbf{J}_{\mathbf{f}}(\bar{\mathbf{x}}, \bar{\mathbf{u}}). \tag{30}$$

In this paper, only the first partial derivative is considered. Since the computation of the Jacobian of the iMTI model is an analytical computation, and the derivatives of the nonlinear functions with respect to time can be precomputed, the Jacobian only needs to be computed once. The operating point is then inserted, maintaining an efficient approach for the linearization.

VII. SIMULATION RESULTS, SMALL-SIGNAL ANALYSIS, AND DISCUSSION

The proposed methodology to model power systems and performing small-signal stability analysis using multilinear models is tested for different scenarios. First, the results of the time-domain simulations of the nonlinear, multilinear, and the obtained linear model are compared. Then the small-signal stability analysis is performed where the generalized eigenvalues of the obtained descriptor state-space model are compared with the eigenvalues of the linear state-space model approximating the nonlinear model at the equilibrium point. The computation time required for the linearization will be compared at the end.

A. COMPARISON OF THE SIMULATION RESULTS

The nonlinear model of the essential network was implemented in Simulink of the Matlab 2023b version. The iMTI model is initialized from the time-domain simulation of the NTI model. Note that the algebraic constraint (8) is used in addition to compute consistent initial conditions for the iMTI model. Similarly, the equilibrium point is obtained from the NTI model, at which the linearization of the iMTI model is performed using (16), and the LDSS model is constructed using (17). The equilibrium point is chosen by inspection.

The `ode23` algorithm for the solution of initial value problems in Simulink [20] is used for the NTI model, which is a single-step solver based on the explicit Runge-Kutta (2,3) pair formula and suitable for moderately stiff problems with crude tolerances. The iMTI model is simulated using the `MTI Toolbox` [24], which uses the `ode15i` solver [20], which is an implicit variable-step, variable-order solver based on the backward differentiation formula. The LDSS is translated into in equivalent model with fewer states, and a non-singular $\mathbf{E}$ matrix using the `isproper` command in Matlab [20], which allows using the `lsim` command to simulate the time response. A maximum step size of $1\cdot10^{-4}$ s, and a relative tolerance of $1 \cdot 10^{-4}$ were chosen for both modeling approaches.

To observe the different behavior of the GFL and the GFM converter in the time-domain simulation, the network is perturbed by typical occurrences in power systems. Firstly, by a small load step to remain in the small-signal regime, and secondly, by a large load step, followed by a voltage step, to excite the model out of the small-signal regime and to trigger a response from the GFM converter on the voltage besides those related to frequency stability. At the beginning of each simulation, the grid-forming and the grid-following model have the same active power reference to share the load equally.

1) SMALL LOAD STEP

A small load reduction of 1 % is triggered at 2.5 s. The iMTI model is initialized and linearized 1 ms before the disturbance. The responses of the grid-forming and grid-following converters are depicted in Fig. 8 and Fig. 9, respectively. The active and reactive power, p, and q, respectively, and the Euclidean norm of the voltage and the current, are plotted in per-unit, and the frequency of the VSM and the PLL in Hz, as well as the angle difference δ between local and global frame, measured in Degrees, are given.

Due to a change in the load, the power flow changes, thereby, altering load angle. In the GFL converter, changes in the load angle perturb the PLL, leading to a small synchronization mismatch in the local voltage and current, which in turn affects the active and reactive power. However, since the GFL converter has a fixed active and reactive power control mode, the active and reactive power settle back to their initial values before the small load step after around 4 ms and 8 ms, respectively.

The response of the GFM converter in Fig. 8 shows the droop dynamic in the VSM control shown (28b), where the active power is reduced to maintain the nominal frequency. The reactive power is perturbed and slightly reduced due to changes in the angle difference, which are affected by the VSM control and cascade down through the reactive power calculation, voltage control, and virtual admittance.

All three, the NTI, the iMTI, and the LDSS model, indicated by the solid blue line, the red dashed, and the yellow dotted line, respectively, show a similar response for both converter control approaches. This is expected since the disturbance with 1 % is very small. Therefore, the LDSS approximates the iMTI model reasonably well. The iMTI model is an exact representation of the NTI model, as shown in Appendix A-B, which can also be seen in the simulation results. Small deviations between the three models during transients are the result of the different numerical solvers, where implicit solvers are generally more suitable when oscillations occur in the model [48]. One drawback of the `ode15i` solver is that it can only simulate square models, i.e., fully determined models with as many equations as unknowns. Therefore, the algebraic constraint (8) is not

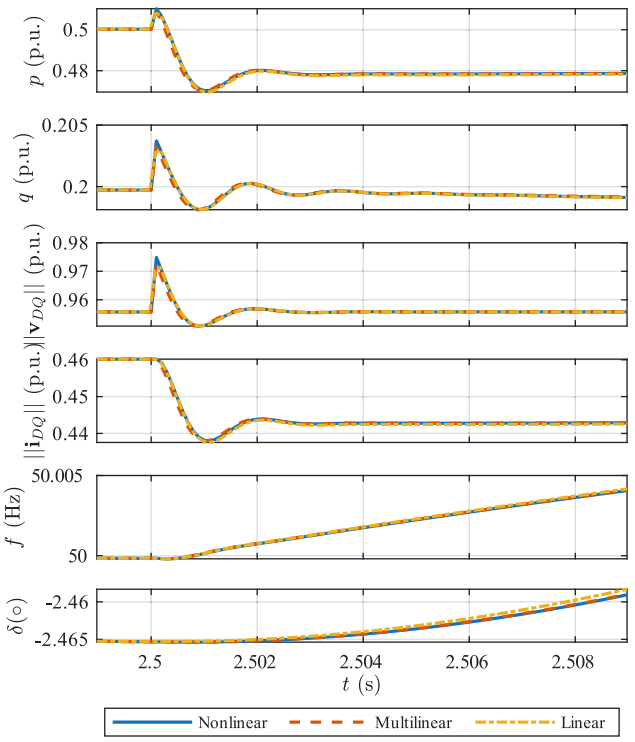


FIGURE 8. Response of the grid-forming converter to a 1 % load reduction at 2.5 s.

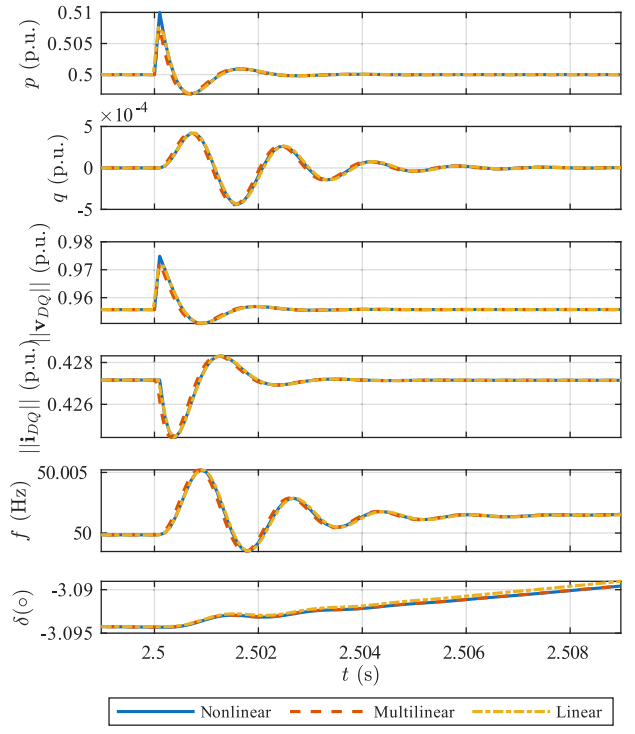


FIGURE 9. Response of the grid-following converter to a 1 % load reduction at 2.5 s.

considered in the numerical integration scheme, which can lead to a numerical drift [49].

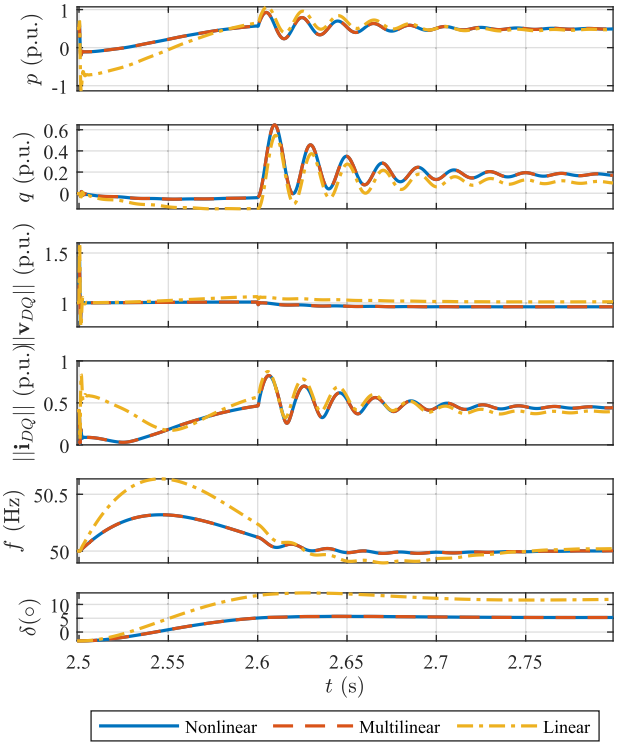


FIGURE 10. Response of the grid-forming converter to a 50 % load reduction at 2.5 s and 9 % voltage drop at 2.6 s.

2) LARGE LOAD STEP

The load reduction is now increased to 50 %. In addition, a voltage drop of 9 % occurs at 2.9 s. The time responses of the GFL and GFM converters are depicted in Fig. 11, and Fig. 10, respectively.

The frequency of the VSM is significantly perturbed by the larger load deviation. The emulated inertia in the VSM slows down the response of the GFM converter, which is why the active power settles back after around 100 ms. The voltage step at 2.6 s leads to an increase in reactive power injection due to the voltage droop in (26b). The oscillation results from the phase lag of the first-order filter in reactive power control and from the virtual admittance. Similar to the previous case of the GFL converter, the load change leads to large initial transients that decay quickly and the converter returns to its initial set-point as shown in Fig. 11. The frequency of the PLL follows the VSM frequency. After the voltage drop, the current injection increases.

The larger disturbances result in larger deviations from the equilibrium point, and consequently, the results of the LDSS model differ more than in the previous case. Besides the significant deviations during transients, the LDSS model also settles for a different steady state. In contrast, the iMTI model and the NTI model show the same behavior as expected.

3) HOPF BIFURCATION

Power systems can experience bifurcations [3], [50], which are nonlinear phenomena [36], i.e., cannot be represented

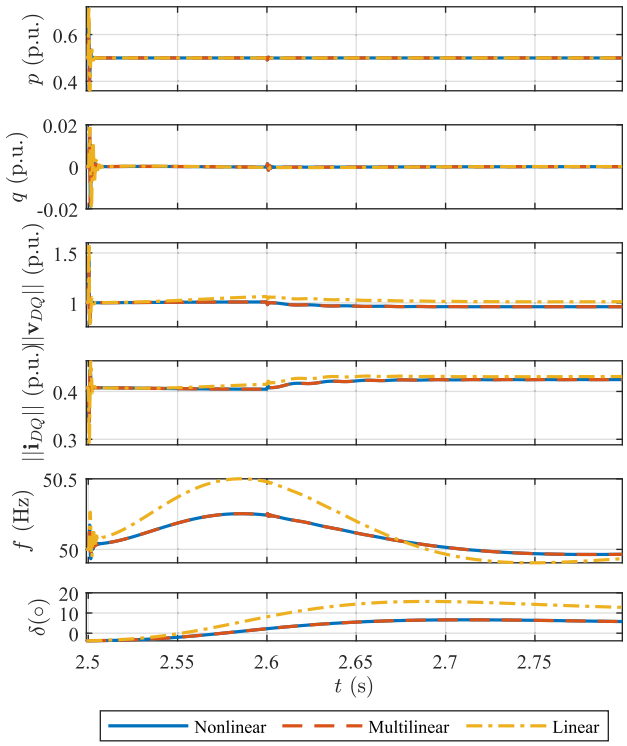


FIGURE 11. Response of the grid-following converter to a 50 % load reduction at 2.5 s and 9 % voltage drop at 2.6 s.

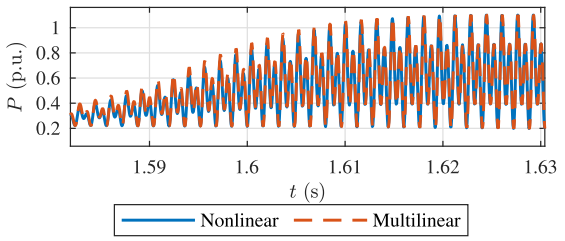


FIGURE 12. Example of a Hopf bifurcation seen in P of the grid-forming converter.

by LSS. The example network is driven on purpose to a point by increasing the active power setpoint to reach the power transfer limit, where a Hopf bifurcation starts to occur, shown in Fig. 12. The iMTI model matches the nonlinear model as expected.

B. SMALL-SIGNAL ANALYSIS

The generalized eigenvalues of the LDSS are compared with the eigenvalues of the LSS of the linearized NTI model to assess the stability of the equilibrium point. The LSS is obtained using the `linearize` command of the `Control System Toolbox`. The `linearize` command linearizes block by block using either an exact analytical linearization, or perturbs the right-hand side of the ODE within the block to linearize it numerically [20], where the default values are used. Both, the eigenvalues, and the generalized eigenvalues, are computed using the `eig` command.

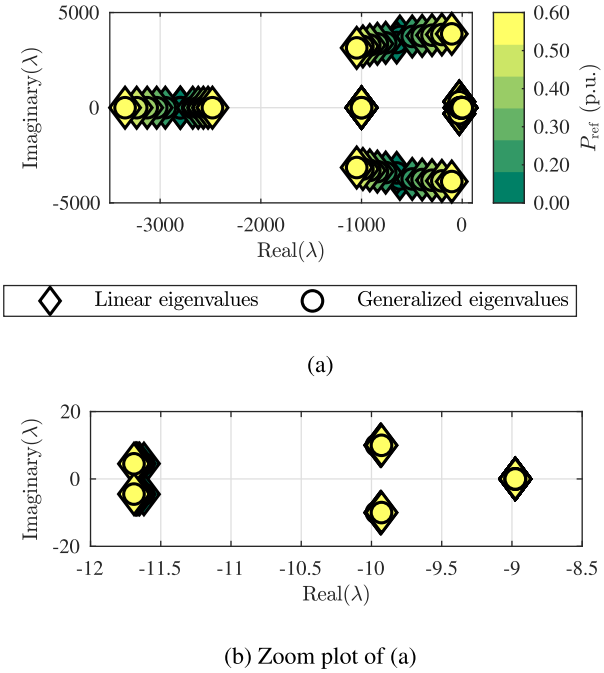


FIGURE 13. Comparison of the generalized eigenvalues of the linearized iMTI model with the eigenvalues of the linearized NTI model.

The models are linearized at 1 ms before the load step, and the results are depicted in Fig. 13. The eigenvalues of the LSS are indicated by the diamond-shaped markers, and the finite generalized eigenvalues by the circles. The colormap indicates the active power set-point, where dark green indicates a low value, which increases with the brightness of the color to yellow. The LDSS has rank($\mathbf{E}$) = 26 finite generalized eigenvalues, with some being zero. The non-zero generalized eigenvalues match the eigenvalues of the LSS, which is expected as mentioned in the previous section. Small deviations occur due to the numerical methods of computing the eigenvalues, as well as the numerical approximation of the Jacobian for parts of the NTI model in contrast to the exact linearization of the iMTI model. In the upper plot of Fig. 13, the increase in the active power set-point results in some eigenvalues moving towards the imaginary axis. As mentioned in the case of the Hopf bifurcation in Sec. VII-A 3), power networks have a power transfer limit. The trajectory of the eigenvalues indicates that, as the active power set-point increases, the network approaches the power transfer limit, leading to a reduced damping of the modes.

The small-signal analysis of the iMTI model comes to the same conclusion as the linearized NTI model, that the equilibrium points are stable, since there are no eigenvalues with positive real part.

C. COMPUTATIONAL ASSESSMENT

The computation time of the proposed linearization method for iMTI models is compared with the linearization using the `Control System Toolbox`™ (CST) from Matlab

TABLE 2. Comparison of the computation time of the linearization (lin) and the eigenvalue computation (eig) for different configuration: a) GFM b) GFL c) GFM+GFL.

	Model properties					CST		MTI Toolbox	
	n	m	$p+q$	R	nnz (%)	lin (ms)	eig (ms)	lin (ms)	eig (ms)
a)	16	6	17	87	3.09	309.7	0.123	0.0133	0.1420
b)	15	6	17	95	3.57	450.34	0.1091	0.0144	0.1221
c)	26	8	32	165	2.03	471.02	0.257	0.0474	0.4776

Simulink. The code in Matlab 2023b [20] was executed on a x64-based laptop with Microsoft Windows 11, an Intel® Core™ i7-1370P processor, and 32 GB RAM. Table 2 compares the computation time for linearization (lin) and eigenvalue calculation (eig), where the model properties indicate the number of states n, the number of inputs m, the number of algebraic and output variables Q, the number of factors of the iMTI model R, and the percentage of non-zero elements of the iMTI model as a CPN1 decomposed tensor. The linearization using the `MTI Toolbox` is significantly faster than the approach using CST, since only the operating point needs to be inserted. The computation of the generalized eigenvalues requires more time than for the computation of eigenvalues in the studied cases. The increase of the size of the iMTI model lead to an increase in computation time as expected. Therefore, for the studied cases, the linearization of the iMTI model was more efficient than the CST.

D. CURRENT LIMITATIONS OF THE PROPOSED METHODOLOGY

The previous subsection demonstrated that restricting the model class to multilinear functions can provide, as shown here, computational advantages, among others. Of course, this restriction comes with a cost, where the presented work is discussed in the following from different angles:

- Limitations of the presented power system model: While all models are an abstraction of the real world, compared to state-of-the-art small-signal stability analysis [40], the work could be extended to include more details, for instance, the dynamics of the DC-link [51], or an accurate network representation beyond a Thévenin-equivalent grid [2], [40].
- Limitations of symbolic multilinearization: Going from an NTI, to an iMTI model, as depicted by the dash-dotted line in Fig. 1, it was shown in Sec. VI that some nonlinear components are multilinear models already and that for other relevant components, e.g., the Park transformation, a suitable change of variables exists to transform the NTI model to an iMTI model, where the recent report [27] presents the multilinear model for a number of power system components, including synchronous generators. However, when considering other plant components, for example, the electrochemical model of an electrolyzer, other nonlinear functions like an exponential function or the natural logarithm appear, for which symbolic multilinearization is not known. Then the NTI model can be approximated using numerical multilinearization [7], [9].
- Remarks on scalability: iMTI models can represent continuous-valued as well as discrete-valued nonlinear dynamics, and thus hybrid, that are multilinear [7]. Sec. III showed that in the step of symbolic multilinearization, the number of state and algebraic variables increases compared to the NTI model. The representation of the iMTI model as CPN1 decomposed tensors breaks the curse of dimensionality, as shown in Sec. II. Table 1 shows that the presented models are very sparse, with a density of 2.03 % for the model containing both converters. However, besides sparsity, a key challenge for large-scale iMTI models is finding a low-rank representation, which is not trivial, but can be seen as a canonical polyadic tensor decomposition, which is known to be NP-hard [52]. Considering converter-dominated power networks, where the number of devices drastically increases compared to legacy power systems, it is the hope of the authors, that as converters will have similar structure and parametrization, the expected repetitive structure can be exploited leading to a low-rank representation.
- Remark on maturity of the tools: When comparing the proposed methodology with existing commercial tools like Matlab, the `MTI-Toolbox` [24] is not as matured and subject to ongoing research.

VIII. CONCLUSION

This paper proposes an efficient approach to perform small-signal power system stability analysis using implicit multilinear time-invariant models, with a focus on converter-dominated networks. It explains how iMTI models are represented by CPN1 decomposed tensors, and how some nonlinear functions can be written as an iMTI model. The iMTI model is an exact representation of the nonlinear model, as the paper shows that the change of variables used to represent trigonometric functions is a Lie-Bläcklund transformation, i.e., the models are equivalent. This equivalence could be seen in both, the time-domain simulation results, and the small-signal stability analysis of an essential power network, where the proposed approach was compared with the nonlinear modeling approach. To enable the small-signal analysis of iMTI models, this paper introduced the efficient linearization of iMTI models, where the iMTI is approximated locally by an LDSS model. From the LDSS model, the generalized eigenvalues were computed to assess the stability of the equilibrium point. A comparison of the computational time also showed that the factorized format of the CPN1 tensor decomposition of the iMTI model enables an efficient linearization compared to the `linearize` command provided by Matlab, with similar degree of accuracy. Fast linearization methods could provide practical benefits for

analysis during real-time operation of transmission systems for decision support, supervisory control, or optimal control, such as model predictive control [13]. The authors want to emphasize, that to their best knowledge, iMTI models are the first model class that enable a structured representation of the studied nonlinear models in power systems, similar to the matrix representation of LSS. The structured representation opens new pathways to system analysis, or model reduction.

Future research will extend the assessment of the proposed iMTI modeling of power systems to larger power networks. The representation as decomposed tensors makes the multilinear modeling class a promising candidate for the modeling and analysis of large-scale low-inertia, converter-dominated networks. Finding lower rank approximations for the multilinear power system models is a key challenge for this. Additionally, the power system models can be extended to include more details, such as saturations, which can be easily represented as iMTI models [7], as shown in a recent report [53] for a simple saturation. Furthermore, other approaches for the stability analysis of iMTI models of power systems beyond small disturbances will be explored in the future with objective of estimating the stability margin.

APPENDIX A EQUIVALENCE OF TRIGONOMETRIC NONLINEARITIES

A. LIE-BLÄCKLUND TRANSFORMATION

Lie-Bläcklund transformations ensure the equivalence of the dynamics of the finite-dimensional model

$$\dot{\boldsymbol{\xi}} = \mathbf{f}(\boldsymbol{\xi}), \tag{31}$$

with vector field $\mathbf{f}$ defined on the smooth manifold $\mathcal{M}$. A Lie-Bläcklund transformation is defined as follows [30], [31].

Definition 1.1: [30], [31] Let $(\mathcal{M}, \mathbf{f})$, and $(\mathcal{N}, \mathbf{h})$, be two models, $\boldsymbol{\sigma} : \mathcal{M} \overset{C^\infty}{\to} \mathcal{N}$, $p \in \mathcal{M}$, and $\mathbf{q} := \boldsymbol{\sigma}(\mathbf{p}) \in \mathcal{N}$. Then,

- if $\boldsymbol{\xi}$ is a trajectory of $(\mathcal{M}, \mathbf{f})$ in a neighborhood of $\mathbf{p}$, then $\boldsymbol{\zeta} := \boldsymbol{\sigma} \circ \boldsymbol{\xi}$ stays in a neighbourhood of q, and

$$\dot{\boldsymbol{\zeta}}(t) = \nabla\boldsymbol{\sigma}\left(\boldsymbol{\xi}(t)\right) \cdot \mathbf{f}\left(\boldsymbol{\xi}(t)\right), \tag{32}$$

  which holds even in infinite dimensions: everything depends only on a finite number of coordinates.
- $\boldsymbol{\sigma}$ is an endogenous transformation if and only if, for any $\boldsymbol{\xi}$ in a neighbourhood of p

$$\nabla\boldsymbol{\sigma}\left(\boldsymbol{\xi}\right) \cdot \mathbf{f}\left(\boldsymbol{\xi}\right) = \mathbf{h}\left(\boldsymbol{\sigma}(\boldsymbol{\xi})\right), \tag{33}$$

  it is said that $\mathbf{f}$ and $\mathbf{h}$ are $\boldsymbol{\sigma}$-related at $(\mathbf{p}, \mathbf{q})$; then $\dot{\boldsymbol{\zeta}} = \mathbf{h}(\boldsymbol{\zeta})$ and $\boldsymbol{\sigma}$ has a smooth inverse $\boldsymbol{\Psi}$. Consequently, $\mathbf{h}$ and $\mathbf{f}$ are Ψ-related.
- $\boldsymbol{\sigma}$ is a Lie-Bläcklund isomorphism if and only if it holds locally that

$$\mathbf{T}\boldsymbol{\sigma}\left(\text{span}(\mathbf{f})\right) = \text{span}\left(\mathbf{h}\right), \tag{34}$$

  and $\boldsymbol{\sigma}$ has smooth inverse $\boldsymbol{\Psi}$ such that $\mathbf{T}\boldsymbol{\sigma}\left(\text{span}(\mathbf{h})\right) = \text{span}(\mathbf{f})$. That is, for $\boldsymbol{\xi}$ in a neighborhood of $\mathbf{p}$, and $\boldsymbol{\zeta}$ in a neighborhood of $\mathbf{q}$, it should hold that

$$\nabla\boldsymbol{\sigma}(\boldsymbol{\xi}) \cdot (\text{span}(\mathbf{f}(\boldsymbol{\xi}))) = \text{span}(\mathbf{h}(\boldsymbol{\sigma}(\boldsymbol{\xi}))), \tag{35}$$

$$\nabla\Psi(\boldsymbol{\zeta}) \cdot (\text{span}(\mathbf{h}(\boldsymbol{\zeta}))) = \text{span}(\mathbf{f}(\Psi(\boldsymbol{\zeta}))). \tag{36}$$

B. PROOF OF EQUIVALENCE OF THE PLL DYNAMICS

The following shows that the transformation (11) used to express the PLL as a multilinear model preserves the dynamics, meaning that the stability assessment is valid for the nonlinear case despite the change of state dimensions.

It needs to be checked if it is an endogenous transformation between the model $(\mathbb{R}^2, \mathbf{f})$ and $(\mathcal{M}, \mathbf{h})$. This is done by checking if the transformation $\boldsymbol{\sigma}$ has a smooth inverse Ψ. Furthermore, from (33) we need to check $\mathbf{h}(\boldsymbol{\sigma}(\mathbf{x}), \mathbf{u}) = \nabla\boldsymbol{\sigma}(\mathbf{x})\mathbf{f}(\mathbf{x}, \mathbf{u})$. The procedure from [54] is followed.

By checking if $g \circ \boldsymbol{\sigma} = 0$, we can check if the transformation $\boldsymbol{\sigma}$ is well-defined. Calculating

$$\begin{aligned} g\left(\boldsymbol{\sigma}(\mathbf{x})\right) &= \sin^2(x_1) + \cos^2(x_1) - 1, \\ &= 1 - 1 = 0, \end{aligned}$$

shows that $\boldsymbol{\sigma}$ is well-defined.

Then we check if $\boldsymbol{\Psi}$ is a smooth inverse to $\boldsymbol{\sigma}$. The inverse transformation is defined in terms of complex variables as

$$\Psi(\mathbf{z}) = \left(\arg(z_1 + iz_2), z_3\right), \tag{37}$$

where i is the imaginary unit. Each argument is checked if it is well-defined and smooth in the following.

The angle $\arg(z_1 + iz_2)$ is well-defined and smooth, if and only if $z_1 + iz_2 \neq 0$. The constraint $g(\mathbf{z}) = z_1^2 + z_2^2 - 1 = 0$ in (8) forces, when $z_1 = 0$, then $z_2 = \pm 1$, and vice versa. The equivalence constraint (8) is smooth and well-defined. Hence, Ψ is well-defined. Furthermore, Ψ is smooth as it is C^k, i.e., k-times differentiable. Replacing $\mathbf{z}$ with the transformation $\boldsymbol{\sigma}(\mathbf{x})$ leads to the original states

$$\begin{aligned} \Psi\left(\boldsymbol{\sigma}(\mathbf{x})\right) &= \left(\arg\left(\cos(x_1) + i\sin(x_1)\right), x_2, u_1, u_2\right), \\ &= (x_1, x_2, u_1, u_2). \end{aligned}$$

Lastly, it needs to be shown that

$$\mathbf{h}\left(\boldsymbol{\sigma}(\mathbf{x})\right) = \nabla\boldsymbol{\sigma}(\mathbf{x}) \cdot \mathbf{f}(\mathbf{x}, \mathbf{u}),$$

where

$$\mathbf{h}\left(\boldsymbol{\sigma}(\mathbf{x})\right) = \begin{pmatrix} -\left(x_2 - k_\text{p}\left(u_1\sin(x_1) + u_2\cos(x_1)\right)\right)\sin(x_1) \\ \left(x_2 - k_\text{p}\left(u_1\sin(x_1) + u_2\cos(x_1)\right)\right)\cos(x_1) \\ -k_\text{i}\left(u_1\sin(x_1) + u_2\cos(x_1)\right) \end{pmatrix},$$

and

$$\nabla\boldsymbol{\sigma}(\mathbf{x}) = \begin{pmatrix} -\sin(x_1) & 0 & 0 & 0 \\ \cos(x_1) & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

and

$$\mathbf{f}(\mathbf{x},\mathbf{u}) = \begin{pmatrix} x_2 - k_\mathrm{p}\left(u_1 \sin(x_1) + u_2 \cos(x_1)\right) \\ -k_\mathrm{i}\left(u_1 \sin(x_1) + u_2 \cos(x_1)\right) \\ 0 \\ 0 \end{pmatrix},$$

which is true.

Thus, the transformation (11) is an endogenous transformation of the Lie-Bläcklund transformation.

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CHRISTOPH KAUFMANN received the B.Sc. degree in industrial engineering and management from the University of Hamburg, in 2016, and the M.Eng. degree in renewable energy systems from HAW Hamburg, in 2019. He is currently pursuing the Ph.D. degree with the Technical University of Catalonia and the Hamburg University of Applied Sciences, on stability analysis of power systems using multilinear models. He is a Research Associate with the Fraunhofer Institute for Wind Energy Systems IWES.



GEORG PANGALOS received the Dipl.-Ing. degree in electrical engineering and the Ph.D. degree in control engineering from Hamburg University of Technology, in 2011 and 2015, respectively. Since 2015, he has been with the Fraunhofer-Gesellschaft zur Förderung der angewandten Forschung e.V., Hamburg. His research interests include renewable energy integration and energy systems transition.



GERWALD LICHTENBERG received the Ph.D. degree in control engineering and the Habilitation degree in systems theory from Hamburg University of Technology, in 1998 and 2012, respectively. He studied physics at Heidelberg University and the University of Hamburg. Since 2013, he has been a Professor of physics and control with Hamburg University of Applied Sciences and the Scientific Director of the Fraunhofer Application Center for Integration of Local Energy Systems. His research interests include multilinear modeling and tensor decomposition applications for control problems.



ORIOL GOMIS-BELLMUNT (Fellow, IEEE) received the degree in industrial engineering and the Ph.D. degree in electrical engineering from the Technical University of Catalonia (UPC), in 2001 and 2007, respectively. He joined Engitrol S.L., in 1999, as a Project Engineer of automation and control. Since 2004, he has been a Professor with the Electrical Engineering Department, UPC, and participates in the CITCEA-UPC Research Group. He has been an ICREA Academia Researcher since 2020 and co-founded the start-up eRoots Analytics in 2022, focusing on modern power systems analysis. His research interests include power electronics, power systems, and renewable energy integration.